Inbrev

Hack Apertus 2026 · Track 2B · Team Apertool

Run locally, the practice inbox never leaves the practice.

Inbrev sorts the inbox of a Swiss GP practice by handling path and deadline, with Apertus, built to run on a box in the practice. Every field it extracts quotes its source.

▶ Watch the 2-min demo

Code on GitHub

solo builder · fictional data only

THE PROBLEM

Before the first patient, there is the inbox.

Lab reports, refill requests, insurer forms, referrals and faxes in German, French and Italian all land in one queue. A medical practice assistant (MPA) reads them one by one.

9.5 h / week

of administration for Swiss GPs and paediatricians in 2025, up from 8.6 h in 2020.

Zeller, Fischer, Hernandez, *Praxis* 2026 · 5th Workforce Study, n = 1,776

54 min / day

that practice-based doctors spend on requests from authorities and insurers alone.

gfs.bern for FMH, 2025 · N = 1,532 doctors

Art. 321

Swiss Criminal Code: professional secrecy binds doctors **and their assistants**. Every message in that inbox is covered.

SCC, SR 311.0 · health data are sensitive personal data under FADP Art. 5

HOW IT WORKS · PURPOSEFUL USE OF AI

Sorted by who handles it and by when. Every field quotes its source.

01 · READ

Mail, forms, faxes

E-mail files and IMAP (tested on a local test server), web forms, fax PDFs via OCR (tested); HIN planned. German, French, Italian and English.

02 · ROUTE

9 handling paths

Report, appointment, refill, insurer, referral, certificate, billing, marketing, other. Plus an administrative deadline and who takes it: MPA, doctor or admin.

03 · PROVE

Quote or drop

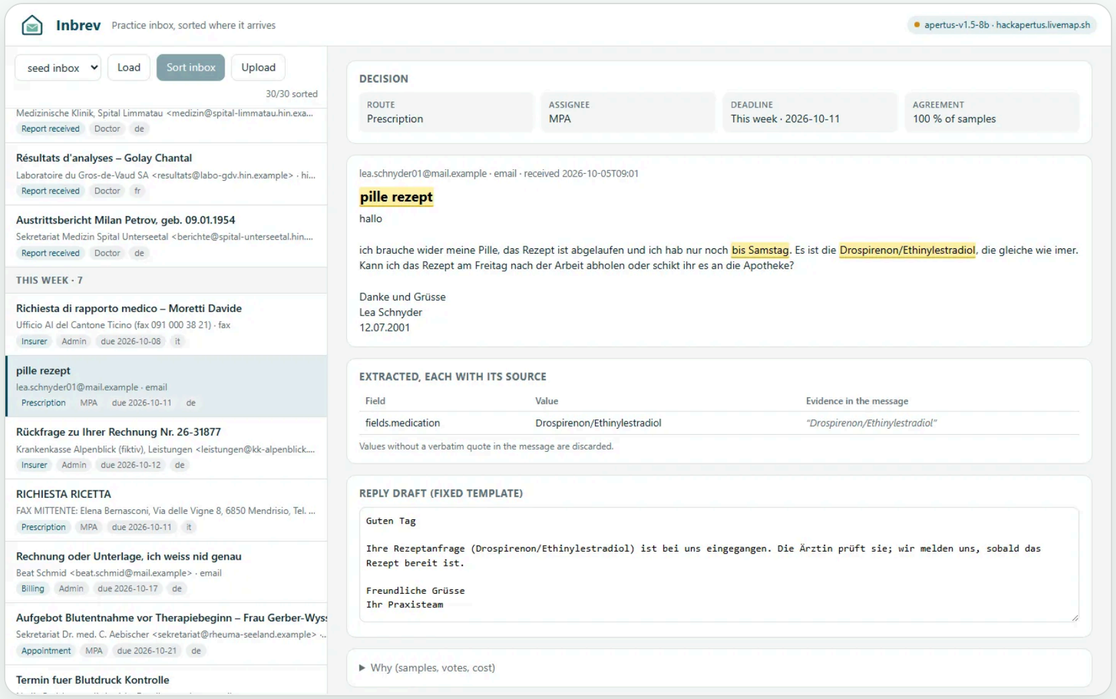
Every field carries the exact words it came from. Code checks that the quote is in the original. If Inbrev is unsure, **a human decides**.

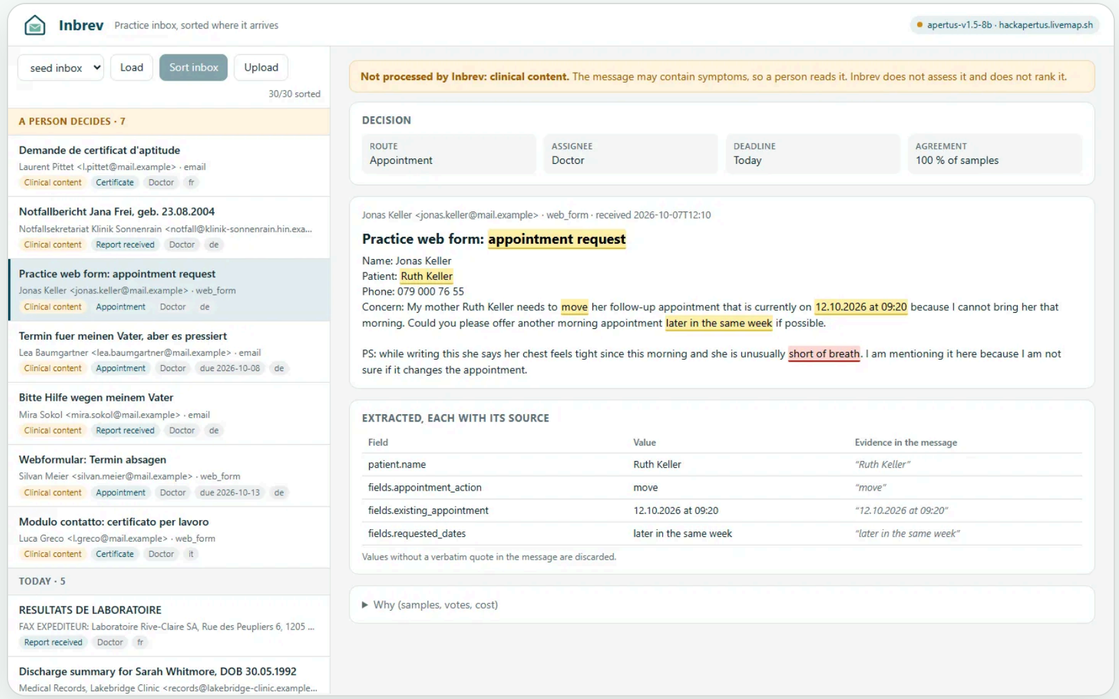
Clinical content A message that may contain symptoms is excluded from automation and goes to a person, like any message Inbrev is unsure about. Inbrev does not judge it and does not rank it.

Apertus reads free text in four languages. Everything code can check (AHV numbers, dates, quotes) is checked by code. Small local models do best on narrowly cut tasks (Xu et al., *JMIR* 2026, CHUV).

DEMO

One click. Sorted by path and deadline; what needs a person goes to a person.





Screenshots of the running app with stored Apertus 8B results on fictional test messages. Left: handled automatically, every value with its quote. Right: shortness of breath mentioned in passing, so a person reads it; no ranking.

ARCHITECTURE · SOVEREIGN DEPLOYABILITY

Pull the cable. Inbrev keeps sorting.

INSIDE THE PRACTICE · NO OUTBOUND CONNECTION AT RUN TIME

Inbox
.eml, HIN mail, web form, fax PDF

›

Parse + OCR
pdfplumber, Tesseract deu / fra / ita

›

Rule extractors
AHV no., dates, received time

›

Apertus v1.5 8B
forced JSON schema, thinking off

›

Quote checker
quote not in original → field dropped

›

3 runs agree?
no → “you decide”

›

Sorted inbox
web UI, JSON, local audit log

On-prem box	Mini PC in the practice; CPU or small GPU
Air-gapped	Model host offline; a mail gateway hands messages in
Swiss cloud	Option for single practices, Swiss hosting
Local Q4	Offline in Docker on CPU: 19 messages / hour; same weights on the laptop GPU: 62.7 s / message

```
Offline run, measured
network practice internal: true
egress dns / tcp / https: FAILED
weights sha256 a037df8d… = HF
30 messages, Q4 on a 28-W laptop CPU
route 24/30 · endpoint 23/30
192 s / message · RAM 6.2 GiB
```

EVALUATION · TECHNICAL RIGOUR

Measured: what leaves automation, and what still runs by itself.

180 synthetic messages, labels not reviewed by a person. Symptom pass chosen on a dev split, measured once here. Challenge set, 44 clinical: 8B 41/44 with p3 (dev for the second check), 70B 44/44, rules 31/44.

Setup	Macro-F1 9 paths	Clinical exclusion recall	Recall “today”	Runs automatically
Keyword rules + lexical net	0.76	0.44 (7/16)	0.76	88 %
TF-IDF + log. regression	0.82	0.13 (2/16)	0.66	99 %
Apertus 8B, one answer, no checks	0.84	0.56 (9/16)	0.70	95 %
Inbrev with Apertus 70B	0.96	0.94 (15/16)	0.94	82 %
Inbrev 8B, before 13 Oct	0.94	0.81 (13/16)	0.88	84 %
Inbrev 8B + symptom pass	**0.94**	**1.00 (16/16)**	**0.94**	**78 %**

Prompt p3, built on a separate dev set. Runs automatically = neither unsure nor flagged. Paired on the 180, 8B beats the rules 32 to 5 ($p = 7 \cdot 10^{-6}$). With the symptom pass 41 % of flags are clinical (16 of 39). Fax pages via OCR: route 50/60 (clean 56/60). Other open models: next slide.

Test set v0: 180 fictional messages

- Labels are fixed before any text is written.
- Text by two other model families (Claude, GPT-5.5), 90 each, plus 42 by an open model (Nemotron) as a control. Apertus never writes or labels test data.
- Two open models (not people) relabelled all 367 test messages blind: κ 0.93 route, 0.97–0.98 clinical. Human review pending.
- Reports written by people: GraSCCo 40/40, PARHAF (French) 37/38 routed correctly.

PURPOSEFUL USE OF AI · MODEL CHOICE

Why Apertus, and what it costs.

Measured: same pipeline, 180 messages

- ✕ Qwen3.5-9B and Ministral 8B: route 177/180, clinical 16/16
- ✕ Apertus 8B: route 170/180, clinical 13/16; 70B 172, 15/16
- ✕ Unnoticed (wrong route + clinical): 8B 9 + 3, Qwen 1 + 0
- ✕ Fix chosen on a dev split, measured once: symptom pass → 8B 8 + 0 at 40/180 to a person (Qwen 1 + 0 at 38). Clinical gap closed, route gap only narrowed (4 + 0 with a rules gate, 69/180)

Why we still choose Apertus

- ✓ Pretrained only on openly available data, robots.txt opt-outs applied retroactively, personal data filtered (arXiv 2509.14233)
- ✓ Published dataset catalogue, EU training-content summary and EU Code of Practice statement (swiss-ai/apertus-data-transparency)
- ✓ Developed in Switzerland; all training artefacts released
- ✓ Local run, quote checks and the unranked safety net work with any open model: no argument for Apertus

A trade-off, not a free choice. Fresh hold-out (146, Gemma 4, frozen before any run): 8B 5 wrong routes run automatically at 40 to a person, Qwen 0 at 24; better prompts alone change nothing. Option for practices with 70B access (e.g. Swiss cloud): 70B second opinion when rules disagree, 3 at 45. A practice that must justify which model reads patient mail can point to Apertus' documented provenance. Not legal advice.

VALUE, COST & SCALABILITY

Swiss francs per 1,000 messages, on hardware the practice owns.

ON-PREM, CPU ONLY · MODEL CALCULATION

Apertus 8B Q4 · mini PC

≈ CHF 12

per 1,000 messages at 60 a day (box, power)

Measured: 192 s / message offline, 1 sample. Assumed: box CHF 700 over 3 years. Up to 450 messages a day; energy alone CHF 0.52

SWISS CLOUD VM

8 vCPU, Swiss provider

CHF 2.39

per 1,000 messages, billed while sorting

CHF 0.0448 / h (Infomaniak list price)

CHF 32 / month if always on · other CPU, not measured

REFERENCE

Swiss token API

CHF 2.59

per 1,000 messages at the listed Apertus 70B price

Tokens measured: 1,398 in, 644 out (8B, p2)

CHF 0.70 / 2.50 per million tokens (Infomaniak)

Prices excl. VAT, 4 Oct 2026. Power: mini PC with the same CPU, 38–39 W (ServeTheHome); ElCom 2026 median.

ASSUMPTION

What it buys (assumption model, not measured): 60 messages a day, 60 s by hand vs 15 s to check a suggestion ≈ 32 MPA minutes a day, net ≈ CHF 350 a month; break-even at 33 s per message. Interviews pending.

INTENDED PURPOSE · IMPLEMENTATION FEASIBILITY

Administrative by default. The clinical mode is research only.

Default mode: Inbrev does

- ✓ Sort by handling path and administrative deadline
- ✓ Extract patient and case fields, each with its quote
- ✓ Hand clinical content to a person, unassessed and unranked
- ✓ Say “you decide” when it is unsure

Default mode: Inbrev never does

- ✕ Rank messages by clinical severity
- ✕ Assess symptoms or suggest care
- ✕ Make a decision a person has not seen
- ✕ Send patient data out of the practice (local mode)

Two modes: the default triage with a safety net (all reported numbers) and an optional clinical mode, off by default and not in the web inbox: a keyword hint on urgency for research, not validated and not certified; it would need classification before any clinical use.
Why the line sits here: MDCG 2019-11 Rev.1 (2025) treats software that automatically reprioritises by clinical criticality based on patient data as medical device software; software that only stores, communicates or searches is not. Switzerland classifies via MepV Art. 15 → EU MDR Annex VIII. Privacy: FADP Art. 5, 7 and 22, and SCC Art. 321.

BUSINESS MODEL & ROADMAP

A module for the software practices already use.

Who uses it	The MPA, first thing in the morning
Who pays	The practice owner or the doctors' network
Price	CHF 150–250 per practice and month HYPOTHESIS
Channel	As a module inside HIN mail or practice software, not one more window
Open core	Hackathon code Apache-2.0. Revenue from operation, connectors, support and updates
Later	Germany and Austria: same workflow, after checking their secrecy and AI Act rules

OCT 2026 · HACK APERTUS
Full evaluation, offline proof, CHF per 1,000 messages; gold set published after human review

NOV – DEC 2026
5–10 interviews with MPAs and practice owners: real volumes, categories, fax share

Q1 2027
Pilot in one group practice on its own box; mail and practice-software connectors

Q2 2027
Written intended purpose, impact assessment, classification review before first sale

Inbrev

Team Apertool · Niklas Wichter, solo builder

Run locally, the practice inbox never leaves the practice.

We are looking for one Swiss group practice to pilot Inbrev on its own hardware in Q1 2027.

Contact: hello@apertool.ch

inbrev.apertool.ch

Demo ▶ 2-min video

Code github.com/NWichter/Hack_Apertus_2B_Inbrev

Data seed and Swiss sets in track_2b/data; gold set after review